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B.Tech.  
Gender: Male  
DOB: 04/05/2004

| Examination   | University | Institute            | Year | CPI / % |
|---------------|------------|----------------------|------|---------|
| Graduation    | IIT Bombay | IIT Bombay           | 2026 |         |
| Intermediate  | CBSE       | Global Public School | 2022 |         |
| Matriculation | CBSE       | Global Public School | 2020 |         |

## SCHOLASTIC ACHIEVEMENTS

- Secured **99.25** All India Percentile in the **JEE Mains** examination among over **1.02 million+** candidates (2022)
- Accomplished a **Top 3%** national rank in **JEE Advanced**, competing against **0.155 million+** candidates (2022)
- Awarded **Branch Change (Top 7%)** in freshmen year owing to exemplary overall academic performance (2023)
- Ranked in **Top 100** out of **0.77 million+** students in **CBSE Aryabhatta Ganit Challenge** (Ajmer Region) (2019)
- Conferred **INSPIRE** scholarship worth **INR 60K** by **GOI** for **top 1%** in **1.4 million+** in CBSE board (2022)
- Grabbed **Top** grades (**AA/AB**) in **13+** courses at IIT Bombay over 6 semesters | **GSSoC** Project Admin (2025)

## PROFESSIONAL EXPERIENCE

**Machine Learning Research Intern** | University of South Carolina - USA (Sept'24 - Present)  
Under the guidance of **Prof. Dr. Amitava Das**, Artificial Intelligence Institute, University of South Carolina (AIISC)

- Investigating **Safety Alignment & Adversarial Robustness** of **Text-to-Image** models through training-free model editing and fine-tuning methods (LoRA, Direct Preference Optimization (**DPO**) and Speculative Decoding)
- Created novel **10K image-prompt pairs benchmark** (safe/attacked prompts) for **Query-Free** attacks manipulating **text encoder (CLIP)** without querying the model, designing **evaluation** metrics inspired by **Pick-a-Pic**

**Data Analyst** | NoQs Digital (June'24 - July'24)  
Received **Letter of Recommendation** and **STAR Intern Award** for exceptional performance from Founder & CEO

- Developed dynamic **Dashboards** using **SQL**, **PowerBI** & **Google Sheets** queries to drive data-centric decisions
- Built web app with **CRUD** capabilities using **Google Apps Script** and Sheets backend managing **3K+** records
- Performed large-scale **data mining**, competitor filtering on **10K+** entries to support **M&A** acquisition strategy

## PUBLICATIONS

**Markdrop** | PyPI - Python Package Index | Documents to Markdown and HTML (Dec'24 - Present)  
*Open-Source Self-Project with 15,000+ installs and 140+ stars. Available using **pip install markdrop***

- Engineered & automated **PDF-to-Markdown** conversion pipeline using **Table Transformers**, **Docling**, **XRef IDs** for structuring **textual & non-textual** data; working on optimized **RAG** with semantic caching & reranking
- Developed** method for generating **in-place** self-explanatory descriptions for non-textual content using **5+ LLMs**
- Enabled modular configuration setup, **CLI** & **Docker** based **deployment**, added one-click table download as **Excel**

## KEY TECHNICAL PROJECTS

**TrueMeasure** | Course Project - Wearable Health Technologies | Guide: Prof. Nirmal Punjabi (Aug'24 - Nov'24)

- Modeled respiratory rate prediction for **wearable smartwatches** by extracting features from ECG and PPG signals
- Achieved **90.44% R<sup>2</sup> Score** using **Random Forest**, outperforming **XGBoost (81.78%)**, **LightGBM (56.98%)**
- Developed 5-class ECG arrhythmia classifier with **97.98%** accuracy using weighted **ensemble** of **LSTM** and **GRU**

**Emoji-Aware Tweet Sentiment Classification** | Self-Project (Aug'25)

- Processed **1.6Mn** tweets with **tokenization**, **stemming**, **lemmatization** & **sentiment mapping** for **960+** emojis
- Vectorized text using **TF-IDF (50K dim)** with **N-Gram** features; combined with emoji scores using weighted score
- Trained **Logistic Regression** using **GridSearch**, **Cross Validation**; achieved **78.49%** accuracy, **0.79 F1 Score**

**Image Caption Generation** | IIT Guwahati | Consulting and Analytics Club (May'24 - Jun'24)

- Build image captioning model using **pre-trained VGG16** as **encoder** and network of **LSTM** layers as the **decoder**
- Fine-tuned the model on the **Flickr8K** dataset; achieved **BLEU-1** & **BLEU-2** scores of **0.57** & **0.34** respectively

**Financial Fraud Detection** | Self-Project (May'25 - Jun'25)

- Visualized data using **t-SNE** to identify **Anomaly** patterns and analyzed feature across **V1-V28 PCA** components
- Built **Gaussian Anomaly Detection** with **power transforms** & **GridSearchCV** optimization to detect fraud
- Developed **Undercomplete Autoencoder** to learn hidden representations of non-fraud transactions to detect fraudulent ones in an unsupervised manner: **F2 Score - 0.962**, **F1 Score - 0.917** and **precision-recall** balance

# OTHER PROJECTS AND COMPETITIONS

**AgentCode** | Agentic Web-Based IDE | Seasons of Code | Web and Coding Club (May'25 - Jul'25)

- Developed web IDE with **React**, **Tailwind CSS**, **Monaco** & **Pyodide** for in-browser live python code execution
- Built **Planner**, **Developer** & **Orchestrator** agents via **LangGraph**, architected backend using **FastAPI** server
- Added **AST-based** internal search, **Merkle Trees** to track changes, used **Tavily** retrieval for external references

**BackTube** | Backend for YouTube & Twitter like Platform | Self-Project (Apr'25 - Jul'25)

- Built **Express** backend with modular **routes**, **controllers** and **middlewares** & tested all endpoints using **Postman**
- Designed data models via **Mongoose**, wrote simple **MongoDB** aggregation pipelines. Implemented **JWT** & **cookie** based **authentication** with refresh/access tokens, bcrypt **password hashing** & secure middleware routing
- Developed two-step media upload using **Multer** & **Cloudinary** with central response, validation, and error handling

**Rail Planner** | Self-Project | Data Structures & Algorithms | **LeetCode: 850+** Qs, **Rating: 1724** (May'25)

- Implemented **Trie** + **DFS** based **autocomplete** for station names and **KMP** for efficient keyword search in reviews
- Used **max heap** to filter reviews by rating. Applied **Quick Sort** on **linked lists** to sort journeys/reviews efficiently
- Modeled stations network as a **Graph** and stored journeys with **Binary Search Trees** for fast lookup and updates

**Deep Generative Models** | Seasons of Code | Web and Coding Club (May'24 - Jul'24)

- Trained **GAN** on **200K+** **CelebA** faces & **Conditional GAN** on **MNIST** for **label-conditioned** image synthesis
- Investigated & compared multiple **VAE** architectures to explore their effectiveness in learning **latent representation**

**Virtual Notepad** | Gesture Based Digit Recognition | Self-Project (May'24 - Jul'24)

- Built Air Canvas using **OpenCV** & **Mediapipe** with gesture-based drawing, erasing, selection via **hand landmarks**
- Achieved **98.6% accuracy** via **Ensemble Voting** over **3** trained **CNNs** for digit classification on binarized canvas

**Electricity Consumption Analysis** | Course Project - CE6001 | Guide: Prof. Arnab Jana (Apr'24)

- Conducted **EDA** on households energy data, clustered them using **K-Means** into **3** based on consumption variance
- Modeled usage via **Polynomial Regression** on environmental & household features, achieved **R<sup>2</sup> Score** of **0.68**

**Qualcomm VisionX** | Ranked 4 (100+ teams) & Won **INR 25.5K** | Techfest Competition (Dec'24)

- Implemented **Reflection Removal** using transformer with cross-scale attention and multi-scale fusion, boosting **PSNR** from 19.23 to 23.61, **SSIM** to 0.83, designed **triple-loss** optimization for transformer-based feature extraction

# COURSES UNDERTAKEN

|         |                                                                                                                                                                                          |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CS & DS | Computer Programming & Utilization, AI and Data Science, Logic for Computer Science, Introduction to MCP, Multi Agent Systems, DevOps, Discrete Structures, Data Structures & Algorithms |
| Maths   | Calculus, Linear Algebra, Differential Equations, Complex Analysis, Numerical Analysis, Statistics                                                                                       |

# POSITIONS OF RESPONSIBILITY

**Operations Coordinator** | **Mood Indigo**, IIT Bombay (May'23 - Jan'24)

*Asia's Largest College Cultural Festival | Footfall: 1,46,000+ | Colleges: 4000+ | Events: 240+ | Viewership: 3+ hours*

- Worked in team of **55** members, responsible for smooth execution of events attended by a crowd of **140K+** people
- **Responsible** for **0.45 Mn+** **Diesel Generator** deals, assistance in **0.13 Mn+** **High-Capacity Crane Contract**
- **Lead** the security and crowd management for the safe conduction of **International Music Festival** & **concerts**

**Mentor** | **IIT Bombay** (Aug'23 - Jul'25)

- **Winter in Data Science (Analytics Club)** | Led **9** students through **CNN**, **Autoencoders**, **GAN**, & **Diffusion**
- **Seasons of Codes (WnCC)** | Guided **10** folks in **Transfer** and **Ensemble Learning** using Inception and ResNet
- **Code Wars (WnCC)** | Mentored **6** freshmen in **Python**, **Git**, and **GitHub** through team-based coding projects
- **FinSearch (Finance Club)** | Mentored **16** students in **Option Pricing** & **MLP/RNN** forecast using log returns

# TECHNICAL SKILLS

|              |                                                                                                       |
|--------------|-------------------------------------------------------------------------------------------------------|
| Languages    | C++, Python, JavaScript, Bash, SQL                                                                    |
| Development  | React, TypeScript, Tailwind, Express, Node.js, Django, Flask, FastAPI, Nginx                          |
| Data Science | NumPy, Pandas, Matplotlib, Scikit-learn, OpenCV, TensorFlow, PyTorch, NLTK                            |
| Softwares    | Git, GitHub, Docker Desktop, MongoDB Compass, Postman, MySQL, L <sup>A</sup> T <sub>E</sub> X AutoCAD |

# EXTRACURRICULAR ACTIVITIES

- Eligible for **INR 0.15 million** equity-free grant from DSSE, IIT Bombay **IDEAS** Program (L1C15) for **OneMail.ai**
- Led **Mood Indigo's Mumbai multicity** with **1000+** participants in over **20+** competitions & events (Dec'23)
- Built financial Q&A **data pipeline** from annual reports at **AfterQuery (YC X25)**, earning **INR 39K** (May'25)
- Completed **McKinsey Forward Program** on **problem-solving** & adaptability, awarded global certificate (Jul'25)
- Completed **Goldman Sachs Virtual Experience**, tackling **10+** tasks strengthening analytical & problem-solving
- Worked with **NSS, IIT Bombay** under **Sustainable Social Development** on **Cloth Collection Campaign**